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Title: Social Media May Cause Emergent SARMs Abuse by Athletes: A Content Quality Analysis of the Most Popular YouTube Videos

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Nikhil Vasireddi, MHA is a medical student pursuing orthopedic sports medicine at Case Western Reserve University School of Medicine. His clinical interests surround the emerging use and effects of novel performance enhancing drugs by athletes. His other research interests include robotic hip arthroplasty approach, outcomes, and complications.

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Jacob G. Calcei, MD is a fellowship-trained orthopedic sports medicine physician and shoulder surgeon at University Hospitals and an assistant professor in the Department of Orthopedic Surgery at Case Western Reserve University School of Medicine. He specializes in the treatment of acute injuries and chronic conditions of the shoulder, elbow, hip, knee and ankle. As an assistant professor in the Department of Orthopedic Surgery at Case Western Reserve University, Dr. Calcei teaches and mentors medical students, residents and fellows. He has authored a number of peer-reviewed scientific papers and book chapters and has research interests in cartilage preservation and restoration and the use of biologics in treating sports

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medicine injuries and early arthritis. He also has a particular interest in injury prevention and finding ways that we can make sports safer while optimizing athlete performance.

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Abstract

Objectives:

Selective Androgen Receptor Modulators (SARMs) social media interest is at an all-time high. The aim of this study is to analyze the: (1) quality; (2) educational content; and (3) reliability of the most relevant YouTube videos on SARMs to explain growing SARMs abuse by recreational and professional athletes.

Methods:

'SARMs' was queried (November 28, 2021) through the YouTube video library. The top 100 videos filtered by relevance were categorized by source, type of content, educational quality by Global Quality Score (GQS), reliability by Journal of American Medicine Association (JAMA) criteria, YouTube tags, attitude towards SARMs use, and whether the video provided specific support on how to use SARMs. For all outcome variables, descriptive statistics and comparison among source types and category types were performed.

Results:

Mean JAMA score was 1.6 ± 0.7 out of 4. Mean GQS score was 2.5 ± 1.1 out of 5. Patient videos were of lower educational quality than athletic trainer videos (GQS: 2.11 ± 0.95 vs. 2.95 ± 1.00, $p < 0.01$), and videos categorized as user experience were of lower educational quality than videos categorized as general SARMs information (GQS: 1.92 ± 0.90 vs. 2.72 ± 1.07, $p < 0.05$). User experience and dosing recommendation videos were statistically significantly more positive in attitude than both general SARMs information and SARMS vs. other PEDs.

Conclusion:

Quality, content, and reliability of SARMs YouTube videos was low. Social media likely causes SARMs abuse through disseminating biased SARMs misinformation. These results serve to educate public health oversight bodies, healthcare providers, and sports team members to better identify signs of SARMs abuse, and promote discussion to discourage SARMs abuse.

Keywords: Performance enhancing drugs, Drug Abuse, Substance use disorder, Selective Androgen Receptor Modulators, YouTube, Social media, Athletes

Data Availability Statement:

The data that support the findings of this study are available from the corresponding author, NV, upon reasonable request.

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Introduction

Selective Androgen Receptor Modulators (SARMs) are a class of performance enhancing drugs (PEDs) that act as androgen receptor ligands that achieve tissue-selective activation of androgenic signaling. The non-steroidal forms of SARMs were discovered in the late 1990s, originally demonstrating potential to treat muscle wasting in patients with cancer, osteoporosis, advanced age, and other chronic illnesses [1]. The potential to slow the functional decline of older individuals using targeted androgen signaling has led to the development of non-steroidal SARMs from nearly all major pharmaceutical companies [2]. Clinical trials, *in vivo,* and *in vitro* studies suggest that SARMs can build muscle mass, much like androgenic-anabolic steroids [3-7]. Athletes and social media influences use this literature to justify SARMs use and promote SARMs on social media. However, SARMs may cause adverse effects, such as liver injury and acute myocarditis, and SARMs abuse should be recognized as a substance use disorder [8-14].

SARMs, like other PEDs, provide unfair advantages in sports activities, and SARMs abuse is rampant at the professional level. Between 2016 and 2019, the World Anti-Doping Agency (WADA) saw a 4-times increase in the number of SARM positive results in failed drug tests for professional sporting events worldwide [15-18]. Although the National College Athletic Association (NCAA) in the USA does not provide data on their drug testing results, they explicitly added SARMs to their banned substances list in 2015 [19]. Many reports of collegiate-related SARMs-use have appeared in the news and social media since then. In a 2005 U.S. Government hearing titled 'Steroid Use in Sports,' the US House of Representatives described a 'trickle-down effect' on youth from professional athletes and teams [20]. Thus, young impressionable athletes are more likely to abuse SARMs if their favorite athletes and social

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media influencers abuse SARMs. In a cross-sectional randomized response technique survey, 2.7% of young male gym-goers (N = 2264, 24 ± 6 years) reported SARMs use, and 9% reported using anabolic androgenic steroids [21]. Illegal substance use is commonly underreported, and Google Search trends analytics indicate that SARMs' interest is rapidly increasing, and at an all-time high [22]. Thus, true athlete and gym-goer SARMs use prevalence is likely higher than previously reported (>2.7%). [21]

In the USA, SARMs are classified as research chemicals that can be legally bought and sold online; however, marketing SARMs for human consumption is illegal by law. Ironically, numerous YouTube videos support consumption and describe different ways to *properly* dose and cycle SARMs, while hiding behind disclaimers of 'for entertainment purposes only.' The top 100 keyword tagged 'SARMs' videos on YouTube sorted by relevance add up to over 14,000,000 views. Hahamyan et al. [2022] noted that prior to the popular video sharing platform TikTok ban of SARMs-related keywords, videos tagged with keyword SARMs totaled more than 115 million views [23]. The paucity of clinical research on *safe* dosing for SARMs and respective side effects in recreational use suggest users and athletes turn to anecdotal evidence from online forums and now YouTube videos for guidance [22, 24].

Alarmingly, YouTube content creators can post videos without review. Thus, YouTube can be a source of widespread and readily available health misinformation, including celebratory/do-it-yourself videos on SARMs use. In fact, over 93% of adults in the United States use the internet, and 8 in 10 online inquiries for health or medical information start at a generic search engine [25, 26]. Many patients/viewers feel that online medical information is the 'same as' or 'better than'

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information from their doctors [27]. Clearly, to understand SARMs abuse, growing social media content must be analyzed.

The aim of this study is to analyze the: (1) quality; (2) educational content; and (3) reliability of the most relevant YouTube videos on SARMs to explain growing SARMs abuse by recreational and professional sports athletes. Since YouTube video content is not subject to review before being published, the authors hypothesized that the video quality and reliability would be low, and the educational content would be limited with a bias towards SARMs use.

Materials and Methods

Search/Analysis Strategy

On November 28th, 2021, a cross-sectional analysis of the YouTube video library was performed. The term 'SARMs' was queried on YouTube. The first 100 videos sorted by relevance were recorded. A web scraper was coded using the Beautiful Soup and Selenium Python libraries to extract characteristics from the videos. Because these extracted characteristics from YouTube are publicly available, it fulfills criteria for Institutional Review Board (IRB) exemption, so approval and consent was not required.

Extracted Characteristics

The following variables were recorded for each video: (1) title; (2) video duration (minutes); (3) number of views; (4) days since upload; (5) channel name; (6) number of comments; (7) video tags (descriptive keywords assigned to the video upon upload); (8) view ratio (views/day); (9) subscriber count; (10) number of likes; (11) number of dislikes; (12) like ratio (number of likes * 100)/(number of likes + number of dislikes); and (13) Video Power Index (VPI). VPI is defined

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as (like ratio * view ratio / 100) and has been characterized in prior literature as a quantitative measure of video popularity [28-30]. Unique case-insensitive tags were recorded for the 100 videos. Analysis was then restricted to tags that appeared in 4+ videos, and similar tags were then combined into relevant categories. These tag categories were then used for frequency analysis and were included in multivariate regressions.

Video Categorization

Two reviewing authors independently reviewed and analyzed all 100 videos. Any discrepancies of subjective measures were resolved by discussion between the two reviewing authors to remain uniform.

Videos were classified based on upload source into the following categories: (1) physician, (2) non-physician, (3) academic, (4) athletic trainer, (5) commercial, and (6) patient. 'Physician' videos were uploaded by a physician with an MD or DO degree; 'nonphysician' videos were uploaded by non-physician healthcare providers; 'academic' videos were uploaded by authors with non-clinical health science credentials and non-clinical authors affiliated with research groups or universities; 'athletic trainer' videos were uploaded by individuals with any athletic training or coaching credentials; commercial videos were uploaded by a company or organization rather than an individual; 'patient' videos were uploaded by SARMs users not previously described, or any other individual discussing SARMs without any qualifications. Upon upload source categorization, no videos were labeled as 'nonphysician' or 'academic' and these source categories were removed from results reporting and further analyses.

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Videos were also categorized based on their content into the following categories: (1) User Experience, (2) Usage Recommendations, (3) SARMs vs. other PEDs, (4) General SARMs Information, and (5) SARMs Purchasing and Sourcing. Although many of the videos overlapped in content, the category selected for each video reflected a significant majority of the content discussed.

Video Quality and Content

Video quality and reliability was measured using four *Journal of the American Medical Association* (JAMA) criteria. A single point was assigned for the presence of each criterion (i.e., Authorship, Attribution, Currency, and Disclosure) creating a JAMA benchmark score between 0 and 4 [31]. Authorship describes whether the authors and/or contributors provide their affiliations and relevant credentials. Attribution defines whether the video properly cites and lists relevant sources. Currency describes whether the dates that the video was posted and updated are indicated. Disclosure seeks to identify whether the channel and/or contributors disclose relevant sponsors, conflicts of interest, or other support. These criteria have been used to define the reliability of online resources in a variety of clinical topics [28-30, 32, 33].

Video educational content was also subjectively measured using the Global Quality Score (GQS). The GQS utilizes a five-point Likert scale to describe how educational or useful a video is for a viewer [34]. A score of one describes the video to be of poor quality and unlikely to be useful for viewer education, while a score of five describes the video to be of excellent quality and flow and highly useful to viewers. Scores of two through four represent a gradient between the extremes. The reviewing authors independently scored each video and averaged the results.

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Specific to the topic of SARMs, two unique content scores were recorded to identify overall sentiment and the extent of educational information provided on these PEDs. The overall attitude scale measures whether the video held a positive (1), neutral (0), or negative (-1) attitude on the use of SARMs. Videos that support SARMs use were scored positive, and videos that discourage SARMs use were scored negative. Videos that discussed the harms and potential benefits of SARMs use, without supporting/discouraging use were scored neutral. The specific support scale utilizes a binary scale to indicate whether a video provides specific support on how to use SARMs *properly*. Specific support includes information on the length of a PED cycle, dosing, post-cycle therapy, and/or including various concomitant supplements. Because SARMs are not approved for human use, it is useful to quantify how many videos attempt to speak positively about their use in addition to providing usage recommendations.

Statistical Analysis

Descriptive statistics were used to quantify video characteristics as well as the quality and reliability scores. All statistical analyses were performed in R using the relevant packages. The level of statistical significance used for all tests was $\alpha = 0.05$. Spearman's rank correlation coefficient was applied to evaluate correlation between parameters. Furthermore, Bartlett's test of homogeneity of variances was used to evaluate the equality of variances between groups. To understand the data on SARMs available through YouTube, differences in attitude, cycle help, and video characteristics between source groups and content groups were investigated.

The GQS was treated as an ordinal variable while both JAMA score and VPI were designated as ratio variables. Differences among groups in mean GQS ranks were tested with the Kruskal-

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Wallis test. Specific group comparison was then performed using pairwise comparison through the Wilcoxon rank sum test with a Benjamini and Hochberg continuity correction for *p*-value adjustment for multiple comparisons to decrease the possibility of false positives. Furthermore, several different ordered logit regression models were used to model GQS. The best regression model was selected using the lowest Akaike information criteria (AIC), and multicollinearity was checked with variance inflation factors (VIFs).

For both JAMA score and VPI, if groups had similar variances, then differences between groups were evaluated using a one-way ANOVA test with type II sum of squares. However, if variances were statistically different between groups, then Welch's ANOVA test was used, and specific group comparison was then performed with the Games-Howell test with Tukey's method for *p*-value adjustment for multiple comparisons to decrease the possibilities of false positives. Finally, to account for heterogeneity, various multivariate regressions with robust standard errors were tested to best predict JAMA score and VPI. The AIC and adjusted R^2 were used to evaluate regression models with the best regression model having both the lowest AIC and the highest adjusted R^2. Multicollinearity was also checked with VIFs.

Results

SARMs Journal of the American Medical Association Scores

The mean ± standard deviation (SD) for the overall JAMA score for YouTube SARMs videos was 1.6 ± 0.7 out of 4 (Table 1). Differences in mean JAMA scores between video source, content, attitude, and specific support groups were not statistically significant. (Table 2, Fig. 1) While various multivariate regressions were tried using video characteristics as predictors to model JAMA score, the best model still had a negative adjusted R^2. Thus, JAMA score was

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likely independent of video characteristics as the addition of video characteristics as predictors resulted in worse model fitting. Additionally, a multivariate regression with dummy variables for the top 20 tag categories yielded no statistically significant predictors of JAMA score.

SARMs Global Quality Scores

The mean ± SD for the overall GQS score for YouTube SARMs videos was 2.5 ± 1.1 out of 5 (Table 1). Mean GQS scores were statistically significantly lower for patient videos than athletic trainer videos ($p < 0.01$) (Table 2). Mean GQS scores were statistically significantly lower for user experience videos than general SARMs information videos ($p < 0.05$) (Table 2). No statistically significant differences in mean GQS ranks among different attitudes and specific support outputs were found. The best ordered logic regression model to predict GQS (AIC = 401.08) included duration, comments, and subscriber count. However, only duration was a statistically significant predictor of GQS score. Duration had an adjusted odds ratio of 1.066 ($p < 0.01$, 95% CI = [1.022,1.089]). Thus, holding comments and subscriber count constant, for every one-minute increase in duration, the odds of having a higher GQS rank increased by 6.6%. Additionally, a multivariate regression with dummy variables for the top 20 tag categories and duration yielded the SARMs tag as a statistically significant predictor of GQS rank. The SARMs tag had an adjusted odds ratio of 0.2514 ($p < 0.05$, 95% CI = [0.066,0.959]). Consequently, after controlling for other tag categories and duration, for a video with a SARMs tag, the odds of having a higher GQS rank is 74.86% lower than a video without a SARMs tag. Furthermore, a SARMs tag was included in 76% of videos (76/100).

SARMs Specific Content Scores

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With respect to attitude, 36% of videos were negative, 24% were neutral, and 40% were positive. There was a statistically significant association between attitude and different source groups. Patient videos tended to be more positive in attitude (78.9% of videos had a non-negative attitude), while commercial videos tended to be more negative (20% of videos had a non-negative attitude) ($p < 0.05$).

There was also an association between attitude and different content groups. Statistically significant differences were found between user experience videos and SARMs vs. other PEDs videos ($p < 0.01$), user experience videos and general SARMs information videos ($p < 0.05$), dosing recommendation videos and SARMs vs. other PEDs videos ($p < 0.01$), and finally, dosing recommendation videos and general SARMs information videos ($p < 0.01$). User experience videos tended to be more positive with 84.6% of videos having a non-negative attitude. Dosing recommendation videos were overwhelmingly positive towards SARMs as 100% of videos had a non-negative attitude with 83.3% of those videos having a positive attitude. General SARMs information videos were balanced in attitude as 43.3% of videos had a non-negative attitude. Lastly, SARMs vs. other PEDs videos were overwhelmingly negative towards SARMs as 11.1% of videos had a non-negative attitude with all those videos being neutral in attitude.

Forty-seven percent of videos provided cycle help. There was significant association between source groups and specific cycle support outputs. The difference in proportion of videos that are cycle help was statistically significant between physician videos (0%) and patient videos (73.7%, $p < 0.01$), athletic trainer videos (40.4%) and patient videos (73.7%, $p < 0.01$), athletic trainer

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videos (40.4%) and commercial videos (0%, $p < 0.05$), and finally, patient videos (73.7%) and commercial videos (0%, $p < 0.01$).

SARMs Video Power Index

The mean ± SD for the overall VPI was 400 ± 885 (Table 1). Differences in VPI between video source, content, attitude, and specific support groups were not statistically significant (Table 2). The best multivariate regression with robust standard errors (AIC = 1604.74, adjusted R^2 = 0.239, $p < 0.01$) using video characteristics to model VPI included comments, duration, subscriber count, tags, and attitude. Only comments were a statistically significant predictor of VPI. After controlling for the other video characteristics, a one-unit increase in comments was associated with a 0.792-unit increase in VPI ($p < 0.05$). Additionally, a multivariate regression with dummy variables for the top 20 tag categories and comments yielded no statistically significant predictors of VPI using robust standard errors.

Furthermore, the Spearman's rank correlation coefficient demonstrated a statistically significant weak positive correlation between JAMA score and GQS ($\rho = 0.2347$, $p < 0.01$). Spearman's rank correlation coefficient between JAMA score and VPI ($\rho = 0.0397$) as well as between GQS and VPI ($\rho = 0.2601$) was both weak and statistically insignificant.

Extracted Video Characteristics

Duration, number of comments, and number of tags were analyzed (Table 1). From the 100 videos, 1 968 unique case-insensitive tags were collected and processed to 33 relevant categories. 86% of videos had tags, and the frequency of the top 20 tag categories and the categorical distribution of tags related to SARMs and its associated compounds were measured

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(Fig. 2). By video tag count, LGD-4033, Ostarine, RAD-140, and Andarine (S4) were the most popular SARMs, while MK677 and Cardarine were the most popular supplements associated with SARMs. The difference in mean duration was statistically significant between physician videos (8.697 minutes) and athletic trainer videos (17.052 minutes, $p < 0.01$) and athletic trainer videos (17.052 minutes) and patient videos (10.032 minutes, $p < 0.01$). The difference in mean number of comments was statistically significant between athletic trainer videos (773.553) and commercial videos (188.600, $p < 0.01$). Lastly, the difference in mean number of tags was statistically significant between athletic trainer videos (15.574) and patient videos (23.500, $p < 0.05$). Finally, there were no statistically significant differences in mean duration, number of comments, and number of tags among different content groups.

Discussion

This is the first study to quantitatively analyze YouTube SARMs videos for content, quality, and reliability. Google Search trends analytics suggest that SARMs' search interest is at an all-time high, which is consistent with the primary finding that mean views/video was 147,994 (total views for 100 videos: 14,799,400) [22]. Mean views and VPI for SARMs videos are higher than nearly all orthopedic and sports medicine YouTube video analyses to date, even despite large sample size (n=100) [28-30, 35-42]. Our findings are consistent with previous studies that demonstrate significant levels of engagement with online medical information [43-45].

The videos analyzed in this study revealed low educational quality and low reliability of information regarding SARMs. Mean JAMA (1.6) and mean GQS (2.5) scores were consistent with previous literature suggesting that health information YouTube videos are of low educational quality and content [28-30, 35-42, 46-48]. Since the VPI differences by source

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categorization, content categorization, and SARMs specific content scores were not statistically significant, SARMs videos were equally popular by assigned labels validating our methods and demonstrating that categories were chosen fairly.

Although JAMA scores were consistently low regardless of source or content categorization, GQS scores had significant differences by source and content categorization indicating certain SARMs videos are of higher educational quality. Videos by athletic trainers are of higher educational quality than patient videos (GQS: 2.947, 2.105, $p < 0.01$), and videos categorized as general SARMs information were of higher educational quality than videos categorized as user experience (GQS: 2.722, 1.923, $p < 0.05$). One explanation is that videos labeled as patient, user experience, and dosing recommendations provide anecdotal evidence from personal SARMs usage and thus present biased views about SARMs usage. This bias is supported by the statistically significant higher positive attitudes about SARMs for videos categorized as Dosing Recommendations and User Experience vs. other content categorizations. This bias is also supported by the attitude differences between sources, since patient videos were significantly more positive in attitude than commercial videos ($p < 0.05$) and had the highest percentage of positive attitudes on SARMs usage (52.6%).

Potential for propagation of SARMs misinformation is supported by biased anecdotal evidence, the high mean views compared to other health information YouTube videos, low educational quality and reliability scores, and the lack of peer-reviewed clinical guidelines for SARMs. Nearly half (47%) of all SARMs videos analyzed provide specific cycle support, which indicates that numerous videos provide consumption advice on illegal supplements without any clinical

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support. Our findings support prior literature indicating that YouTube is a source of significant biased and misleading health information [49]. Zeydanli et al found that 42% of YouTube videos on Keratoconus were classified as either misleading or patient testimonials [33]. Li et al. found that 19.5% of COVID-19 YouTube videos contained nonfactual information, totaling 62,042,609 views [32]. Through tags and its data mining algorithms, YouTube recommends similar content to viewers, propagating SARMs misinformation [50].

YouTube video tags inform video classifications, hierarchies, and searchability throughout the YouTube algorithm. The tagged SARMs and associated supplements are likely the most recreationally abused, thus future survey-based studies can poll for specific SARM use. Tagging patterns support previous health information YouTube video analyses, suggesting that video views and popularity are prioritized over providing reliable and high educational quality SARMs information [33].

Given that YouTube is a powerful educational and entertainment platform, clinicians and sports teams should be aware of misinformation surrounding PEDs and can look to direct patients and athletes to higher quality literature and properly informed videos. To discourage future SARMs abuse, clinicians and sports team members must discuss the potential harms of SARMs abuse and abundant social media misinformation [23, 51]. Public health oversight bodies can provide evidence-based statements and regulatory documents, especially through social media, to raise awareness, promote discourse, and discourage abuse.

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This study is bound by several limitations. Only 100 SARMs videos were identified for inclusion. This limits the generalizability of results, making the overall content/quality of SARMs YouTube videos difficult to ascertain. However, evidence suggests that only the first few pages of search results are consumed by viewers, so our analysis represents typical search patterns [52]. The lack of high quality SARMs information makes it difficult to assess the validity of information presented in SARMs videos. Despite limiting interobserver variability as described in Methods, intraobserver variability was not quantified. However, GQS scoring, and JAMA benchmark criteria have previously established interobserver and intraobserver reliability [31, 50]. Additionally, the internet, especially social media, is a constantly growing network, and the query period was limited. Performing the YouTube video library search over 3 or more time periods may reveal changes in search results over time.

Finally, YouTube is not the only social media source of information for SARMs. Google Search trends analysis suggests that SARMs information is found all over the internet. Online forums such as Reddit have communities on SARMs consumption and purchasing with ~50,000 members. Several accounts on Instagram, TikTok, and Facebook promote SARMs use and provide online discounts. Content quality analysis on SARMs information can be performed across multiple social media platforms to better understand the emerging use of SARMs as a performance enhancing drug.

Conclusion/Summary

SARMs abuse is undoubtedly growing and likely caused by biased and low educational quality social media misinformation. YouTube SARMs video content, reliability, and educational quality were poor regardless of source and content categorizations as evidenced by the low mean

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JAMA and GQS scores. Due to the lack of clinical guidelines, safety profiles, and peer-reviewed literature on SARMs use, there exists an abundance of anecdotal SARMs information. Social media can be a compelling source of education, but misinformation is abundant. Public health oversight bodies, clinicians and sports team members must be aware of the social media misinformation surrounding PEDs to promote discussion and prevent abuse.

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Disclosure Statement

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Table 1: Video Characteristics

Characteristic	Mean ± SD	Minimum	Median	Maximum
Video Duration (min)	13.7 ± 8.9	1.9	12	57
Views	147,994 ± 185,267	1,273	82,428	1,161,541
Days since Upload	667 ± 445	7	672	1,975
View Ratio	584 ± 638	2.1	160	6,381
Comments	18 ± 12	8	426	3,040
Tags	3,351 ± 4,462	0	21	43
Dislikes	156 ± 253	0	1,802	23,395
Like Ratio	93 ± 8.2	0	64	1,314
Video Power Index	400 ± 885	50	96	99
JAMA Score	1.6 ± 0.7	1	1	3
Global Quality Score	2.5 ± 1.1	1	2.5	4.5

Like Ratio and Video Power Index excluded videos with 0 likes and 0 dislikes

JAMA = Journal of American Medical Association

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Table 2: Video Quality, Reliability, and Popularity Scores

Grouping Variables (*N*)	JAMAS Mean ± SD (Median)	GQS Mean ± SD (Median)	VPI Mean ± SD (Median)
Video Source			
Physician (5)	1.40 ± 0.55 (1)	2.20 ± 1.04 (2)	404 ± 621 (197)
Athletic Trainer (47)	1.53 ± 0.72 (1)	2.95 ± 1.00 (3)	579 ± 1,197 (223)
Patient (38)	1.63 ± 0.63 (2)	2.11 ± 0.95 (2)	251 ± 405 (64)
Commercial (10)	1.70 ± 0.95 (1)	2.35 ± 1.25 (1.75)	113 ± 174 (25)
Video Content			
User Experience (26)	1.69 ± 0.68 (2)	1.92 ± 0.9 (1.75)	257 ± 347 (141)
Dosing Recommendations (12)	1.33 ± 0.49 (1)	2.83 ± 1.05 (3)	203 ± 224 (132)
SARMs vs. other PEDs (9)	1.56 ± 0.73 (1)	2.56 ± 0.98 (2.5)	275 ± 429 (67)
General SARMs information (45)	1.51 ± 0.73 (1)	2.72 ± 1.07 (2.5)	598 ± 1,246 (174)
SARMs Purchasing and Sourcing (8)	2.00 ± 0.76 (2)	2.94 ± 1.12 (3.25)	174 ± 213 (118)

JAMAS = Journal of Medical Association (JAMA) Score; GQS = Global Quality Score; VPI = Video Power Index

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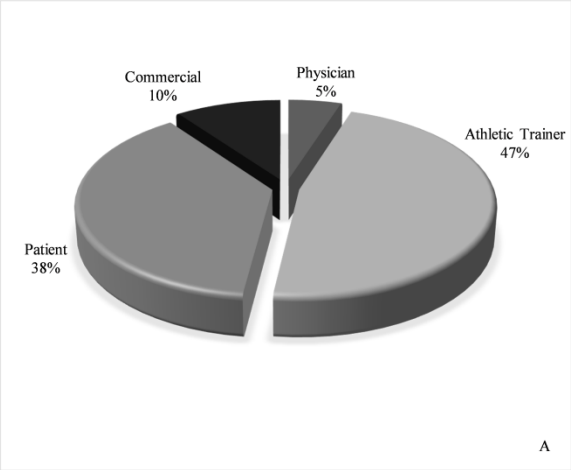
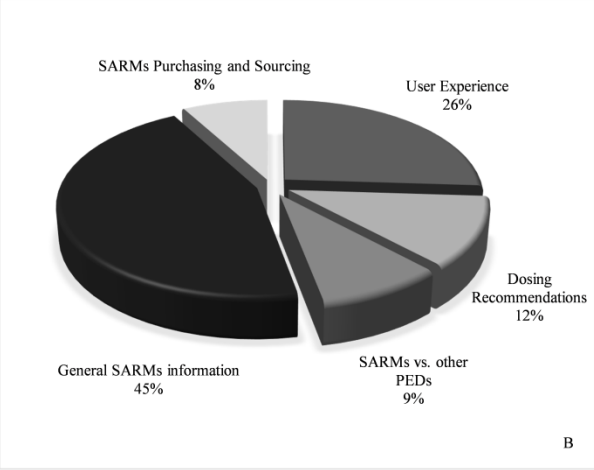
Videos by Category





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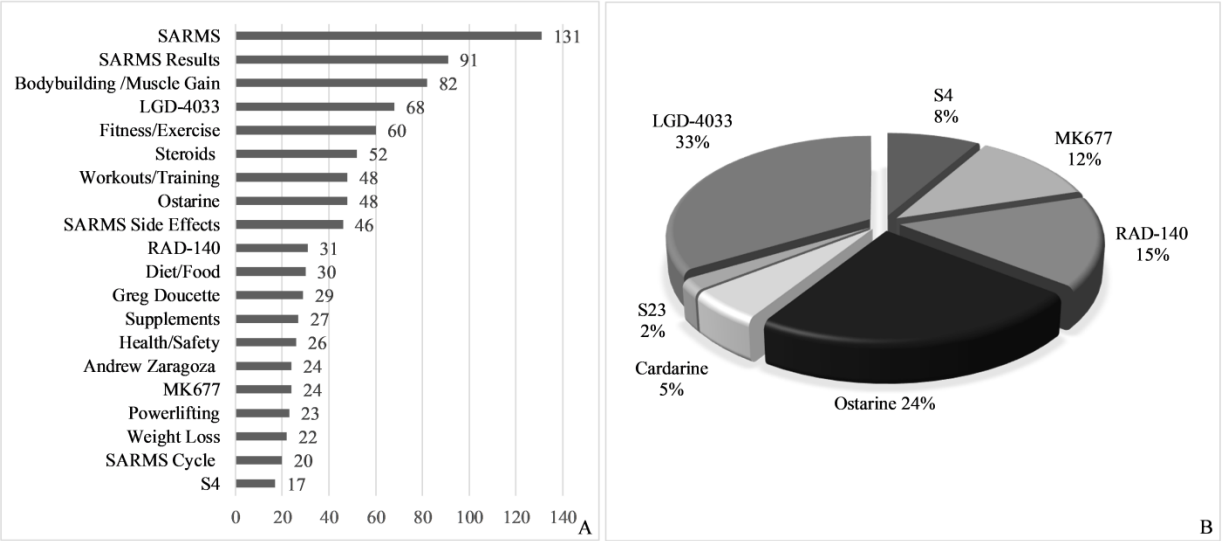
Most Common Tags



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